6) The DBA maintains data accuracy and consistency by enforcing primary key, entrants is consistance between related tables

7) DBA manages database storage capacity transaction log upgrading software maintains & disk space

8) DBA identifies and resolves database error, connection failure, deadlocks corruption, performance problem

9) use and privilege management DBA excats modifies disable database user accounts

10) The DBA ensure database availability through maintain replication, clusters and failure mechanism, where appropriate.

Database Administrator

① A Database administrator is responsible for installing, configuring, securing, maintaining and monitoring an organization database system.

② The DBA helps design, create and manage relationship, database constraints and design suitable database software configuration required

③ The DBA protects data from unauthorized access by implementing authentication, authorization encryption and auditing

④ The DBA schedules and verifies regular database backup recovery procedures are developed

⑤ The DBA makes query execution time CPU usage memory concerned disk

E E DIAGRAM

STAFF

Doctor

Nurse

ADMIN

PATIENT

1 : N

Medical Record

Appointment

1 : N

Admission

1 : N

Hospital Management System

1. Major entities

The Major entities are staff, Doctor, Nurse, Admin, Department, Appointment

2. Staff specialisation and attributes

Staff
- Staff-id
- Name
- Address
- phone
- salary

Doctor
- Staff-id
- Specialisation
- License Number

Department - Cardinality

Dept - Doctor 1:N

Patient - Appoint 1:N

Doctor - Appoint 1:N

Patient - Medical-record 1:N

3. Weak entity and Identifying

Medical record is each entity

Partial Key - Record-Date

Identifying relationship - HAS - RECORD

Hospital Management System

1. Major entities

The Major entities are staff, Doctor, Nurse, Admin, Department, Appointment

2-4 staff specialisation and attributes

Staff	Doctor
Staff-id	* staff-id
Name	* specialisation
Address	* License Number
Phone	
salary	

Department - Cardinality

Dept - Doctor 1:N

Patient - Appoint 1:N

Doctor - Appoint 1:N

Patient - Medical-record 1:N

5-7 weak entity and Identifying

Medical record is each entity

Partial key - Record-Rate

Identifying relationship - HAS - RECORD

6. specialization constrains

Y. Disjoint -> only belongs three subtypes

X total -> Every product belong to three subtypo

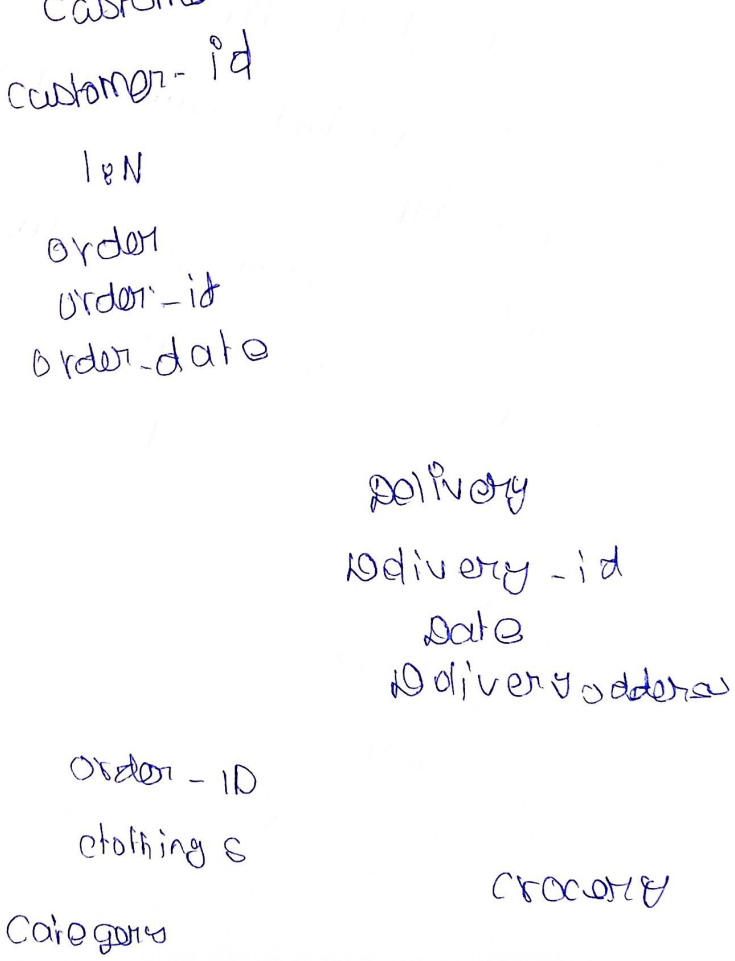
7. 8 Relationship and cardinality

Customer - order -> 1 : N

order - payment -> 1 : 1

order - deliver -> 1 : 1

9. EER DIAGRAM



E - commerce website

① Major Entities

* Customer, Products, category, order, Payment and Delivery

② Attributes and Primary key

* Customer : Customer-ID
* Product : Product-ID
* category: category-ID
* Delivery: Delivery -ID

③ order - Product Relationship

* order and Product have many to many relationship

The attributes Quanty and unit price are stored in order - Item

④ Product is superType

Subtypes are

+ Electronics
* clothing
* Grocery

ER diagram

CUSTOMER

Customer - ID

Customer - Name

Address

Loan

Load - ID

loan - Type

Interest Rate

Account

Account - NO

Account Type

Balance

TRANSACTION

Transaction ID

Date

Type

BRANCH

Branch - ID

Branch - Name

Caream

EMPLOYEE

ID

Name

salary

① The major entities are customer, Account, Branch Employee, Transaction Loan, Loan-Payment

② & ③

Attributes and Primary keys

customer: customer-ID, customer name, Address, Phone, Email

Account: Account-NO

Branch: Branch-ID

Employee: Employee-ID

Transaction: transaction-ID

Loan-payment: payment ID

④ & ⑤

customer-Account 1:N: one customer may hold multiple Accounts

* Branch-Accounts
* Account Transcation
* customer transaction
* Loan Payment

Name : D. Charitha
Subject : DBMS
Reg no : 192126020
Date : 16/10/2024
COURSE : CSA05

ASSESSMENT - I

① Vechicle Management System - EER Diagram

Identify the Supertype and subtypes

Supertype : Vechicle

Subtype : car and truck

Vehicle is the supertype that contains common attributes car and truck are specialised entities that inherit the common attributes of vehicle through an IS A relationship

ⓑ common attributes of vehicle :-

1. Vehicle - ID -> primary key
2. Registration - Number
3. Manfacik
4. Model
5. Year
6. Purchase - Price

specialised Attributes of car and Truck

car Attributes :-

- No. of. Door
- No. of Fuel-type
- Seating - capacity

Truck Attributes

- Load - capacity
- Truck - Type
- No. of Axles